

SMART INDIA HACKATHON 2026



TITLE PAGE · SIGHTLINE

Problem Statement ID – SIH26171

Problem Statement Title – On-device Visual Perception
for Light-weight Browser Agents

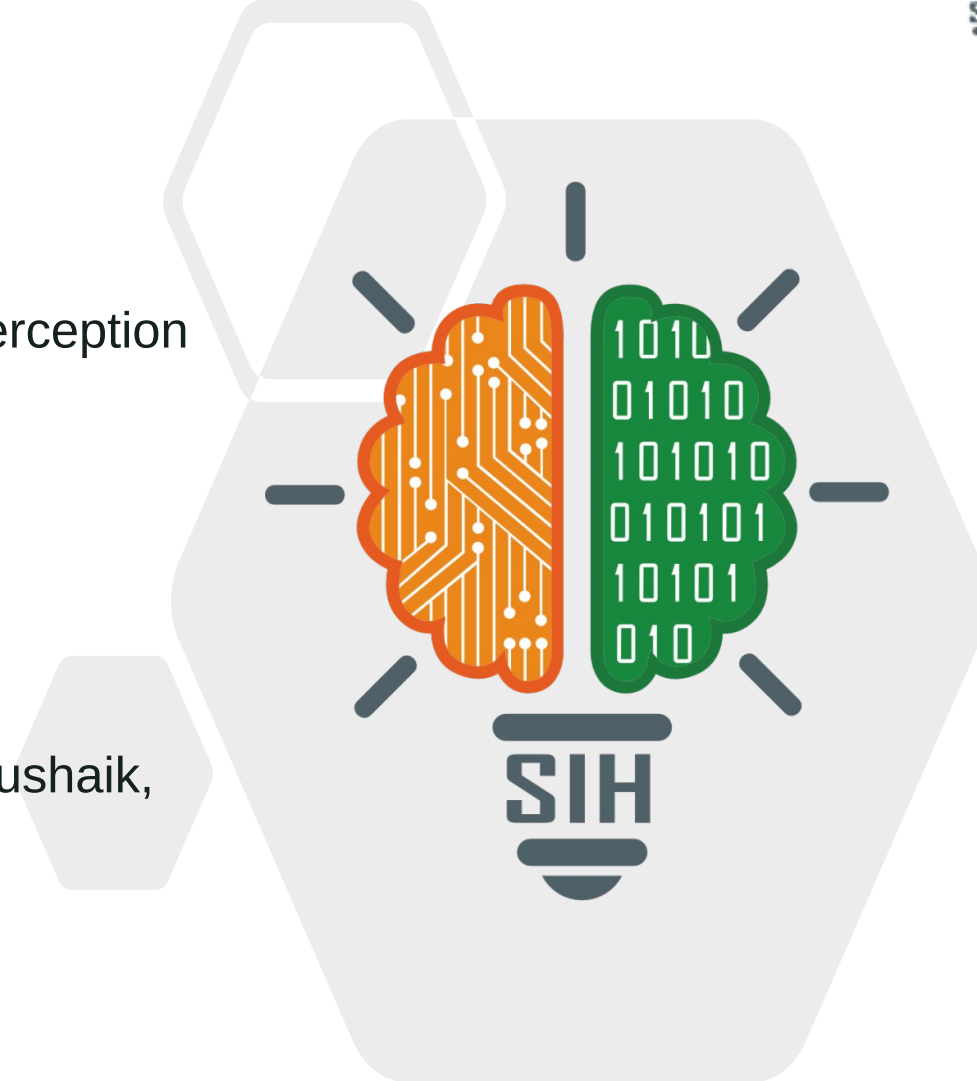
Theme – Smart Automation

PS Category – Software

Team ID – pending registration detail

Team Name (Registered on portal) – pending

RV University · Gopreet, Hiranmayi, Varun, Koushaik,
Francis, Niharika



Proposed Solution (Describe your Idea/Solution/Prototype)

Keep the screen local. Send a protected layout. Review every action.

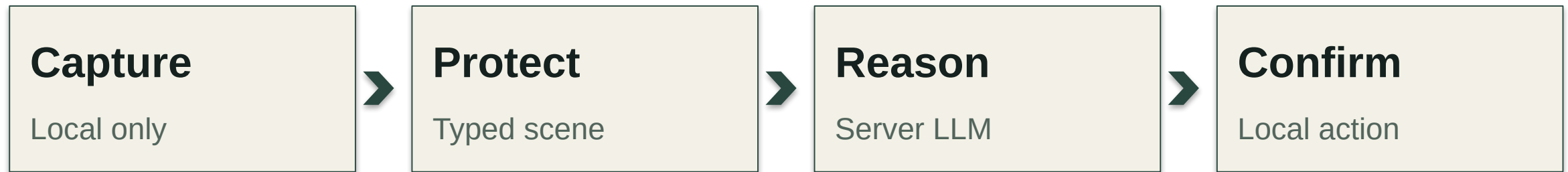
Detailed explanation of the proposed solution	How it addresses the problem	Innovation and uniqueness of the solution
Local face CV + DOM bounds → opaque regions and approved controls → Qwen2.5 → reviewed action.	A strict pixel-free request excludes source text, images, values and URLs before server reasoning.	Inspectable egress + expiring action references. Integration difference; no first-ever privacy claim.

TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)

JavaScript / extension · UltraFace RFB-320 + ONNX WASM · Node.js / Zod
Ollama + Qwen2.5:7B · SQLite audit counts

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Observed: Pending → Review → Request ready for review (synthetic fixture).

Current LLM reads protected geometry/labels, not PNG pixels; native extension QA pending.

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FEASIBILITY AND VIABILITY

Analysis of the feasibility of the idea

Local browser + Qwen workflow observed; 37 recorded Node tests pass.
Single desktop observation: capture 92 ms; face inference 9.6 ms.

Potential challenges and risks

Real model step: 3,029 ms; full-flow <200 ms target fails.
Face-only CV; context removal costs utility; no dataset PII accuracy yet.

Strategies for overcoming these challenges

Measure detection/utility on labeled held-out pages; profile cold/warm costs.
Fail closed on CV errors; explicit action review; expiry and schema checks.

Potential impact on the target audience

Help users complete browser tasks with less raw screen exposure.

Benefits of the solution (social, economic, environmental, etc.)

User control

Inspect server-bound context; approve a bounded action.

Deployable locally

Existing laptop + local model; offline operation after assets are installed.

Measure the trade-off

PII coverage vs useful context; task success; latency and client memory.

Expected benefits, not achieved impact. No user study, savings or energy reduction is claimed.

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RESEARCH AND REFERENCES

Details / Links of the reference and research work

Official SIH26171 · problem and 25/20/20/20/15 weights

<https://www.sih.gov.in/sih2026PS>

WebPII / WebRedact · visual PII benchmark (2026 preprint)

<https://arxiv.org/abs/2603.17357>

PrivWeb · local anonymization and user control (2025 preprint)

<https://arxiv.org/abs/2509.11939>

Available but Invisible · typed placeholders (2026 preprint)

<https://arxiv.org/abs/2602.10139>

UltraFace · MIT model; ORT Web · packaged WASM runtime

<https://github.com/Linzaer/Ultra-Light-Fast-Generic-Face-Detector-1MB>

Prototype evidence: Benchmarks/results/prototype-v01-browser.json · model hashes/licenses in Prototype/models/manifest.json

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